An O(ND) Difference Algorithm and Its Variations

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Abstract (1/4)

The problems of finding a longest common subsequence of two sequences A and B and a shortest edit script for transforming A into B have long been known to be dual problems.

Abstract (2/4)

In this paper, they are shown to be equivalent to finding a Shortest/longest path in an edit graph. Using this perspective, a simple O(ND) time and space algorithm is developed where N is the sum of the lengths of A and B and D is the size of the minimum edit script for A and B.

Abstract (3/4)

The algorithm performs well when differences are small (sequences are similar) and is consequently fast in typical applications. The algorithm is shown to have $O(N + D^2)$ expected-time performance under a basic stochastic model.

Abstract (4/4)

A refinement of the algorithm requires only O(N) space, and the use of suffix trees leads to an O(NlgN + D^2) time variation.

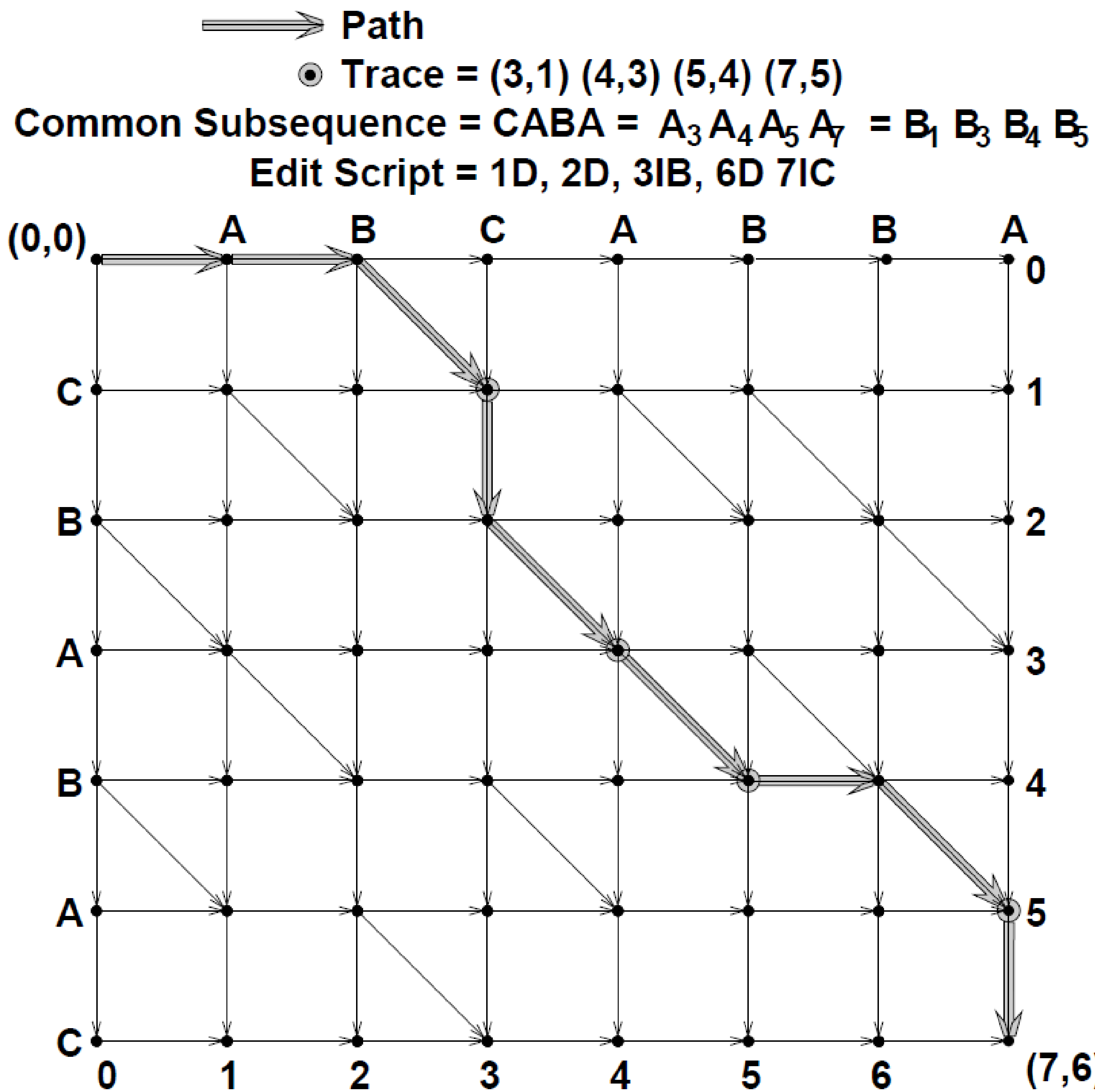
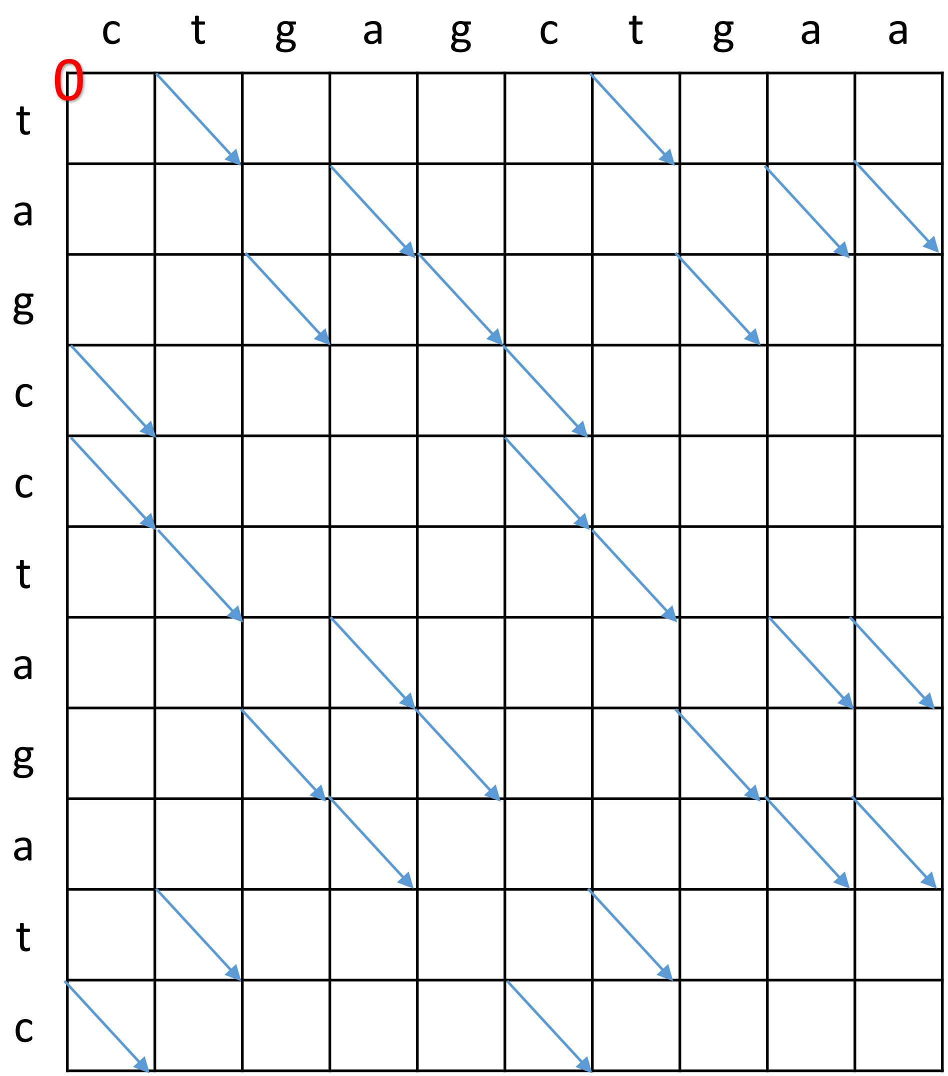
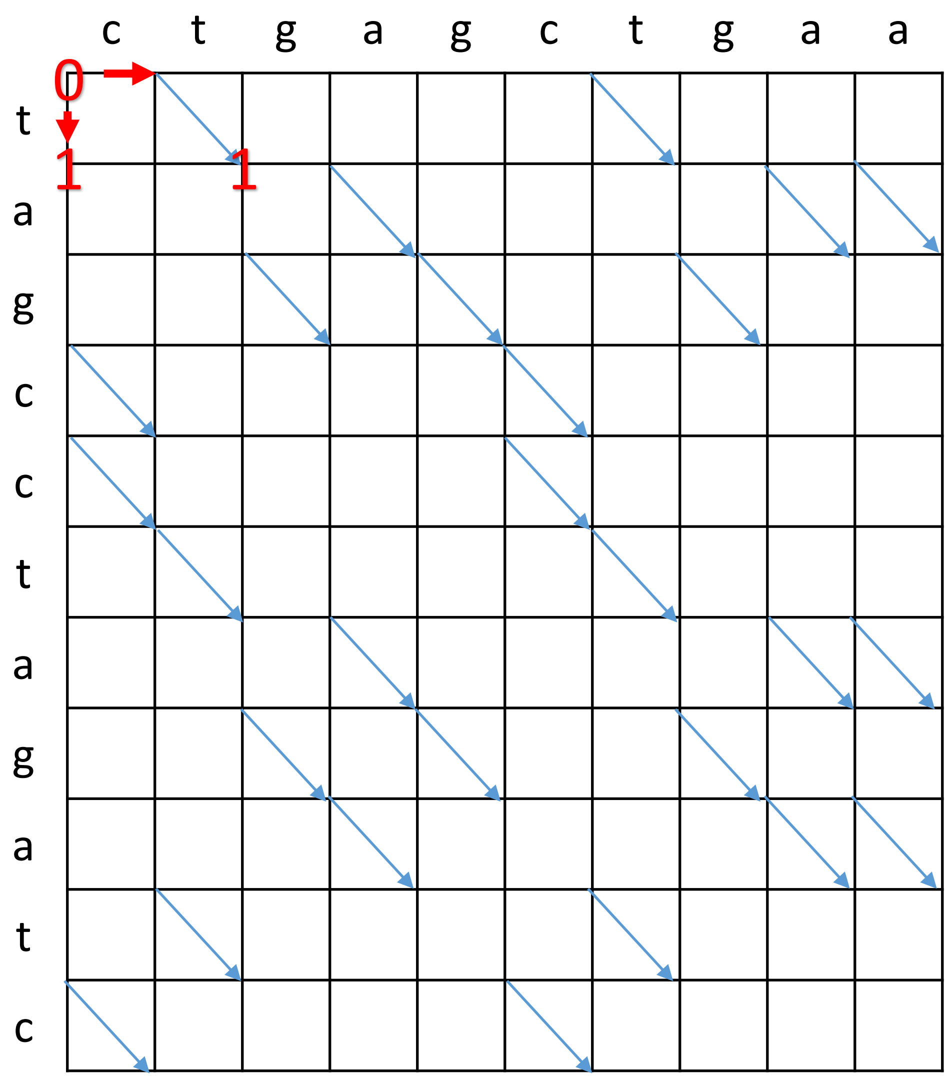
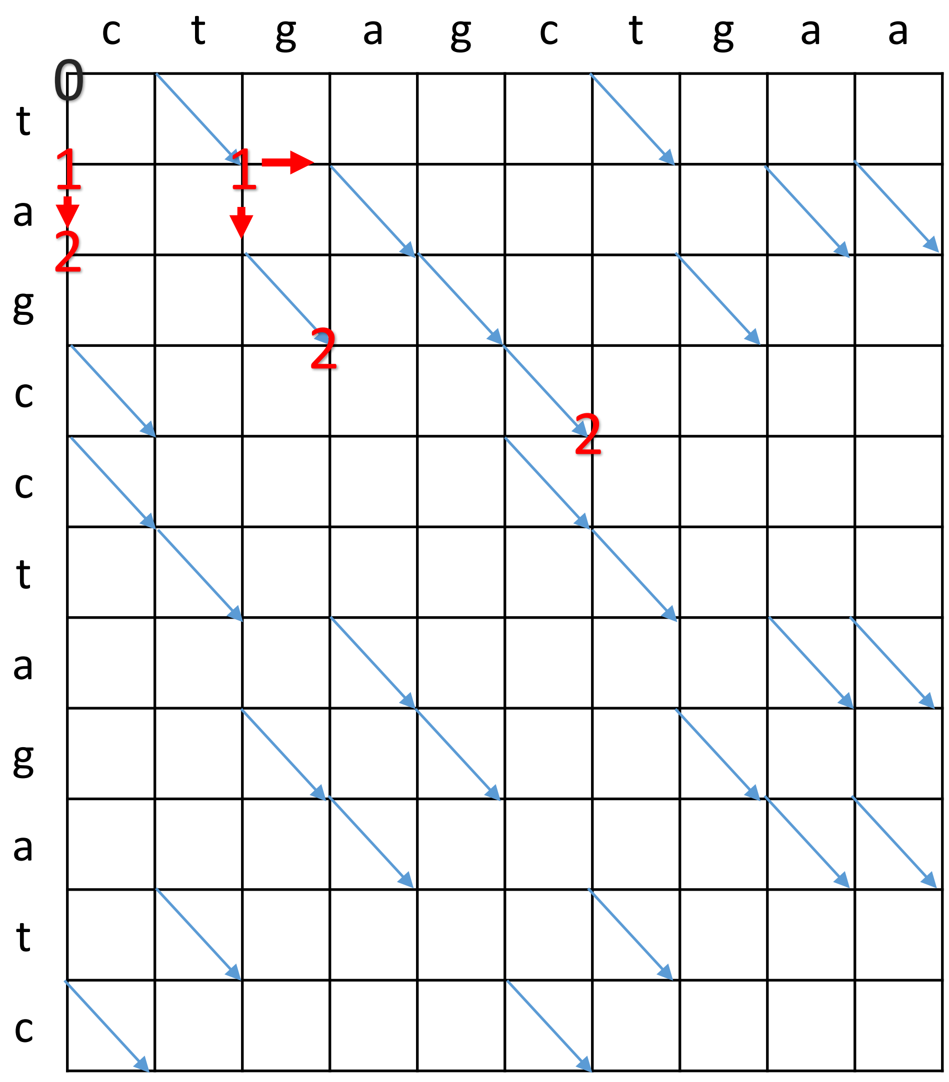


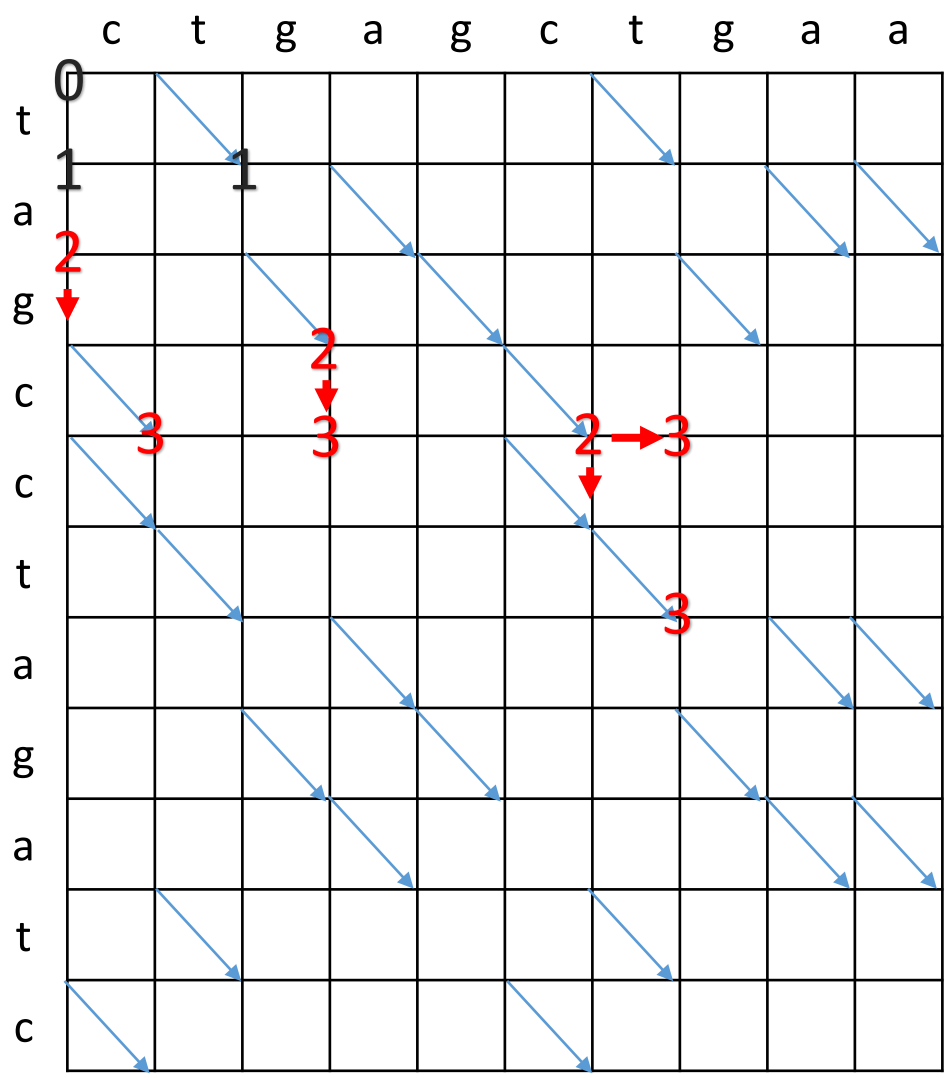
Fig. 1. An edit graph



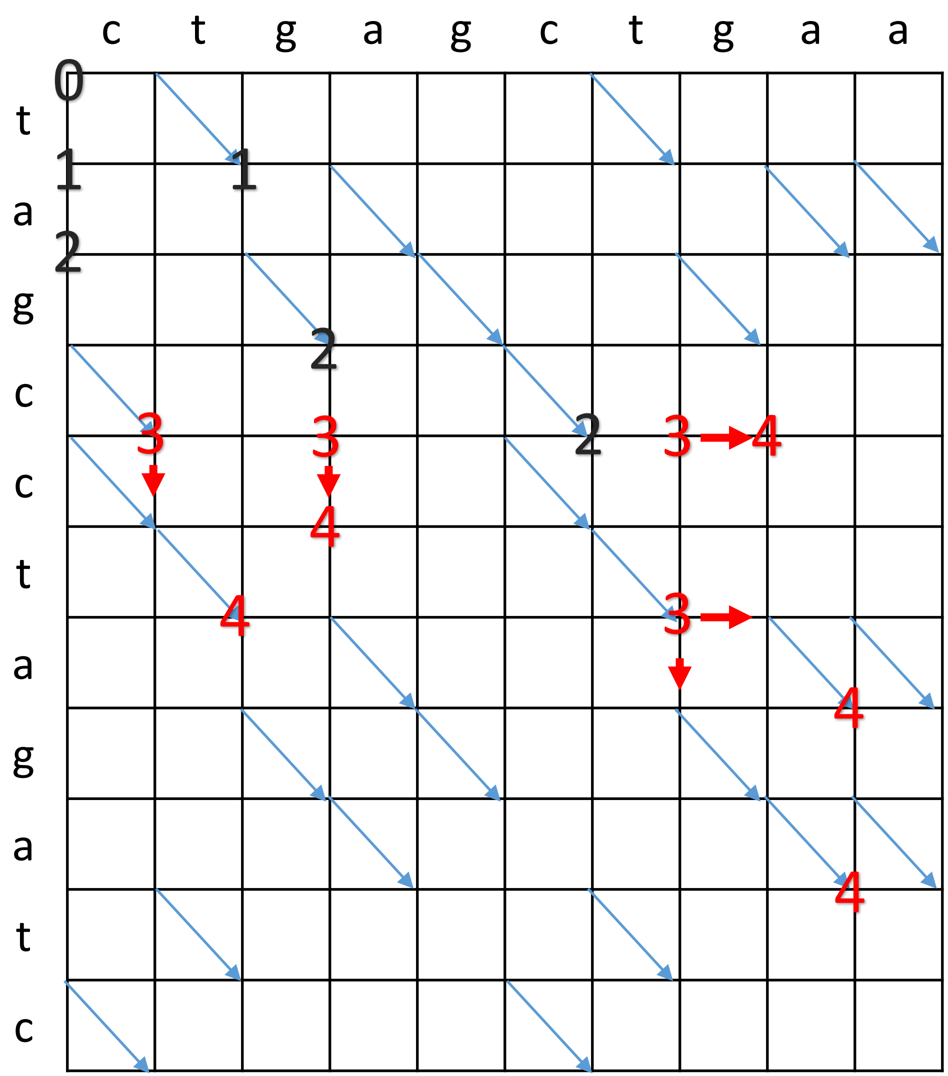
c t g a g c t g a a
0
t
a
g
c
c
t
a
g
a
t
c

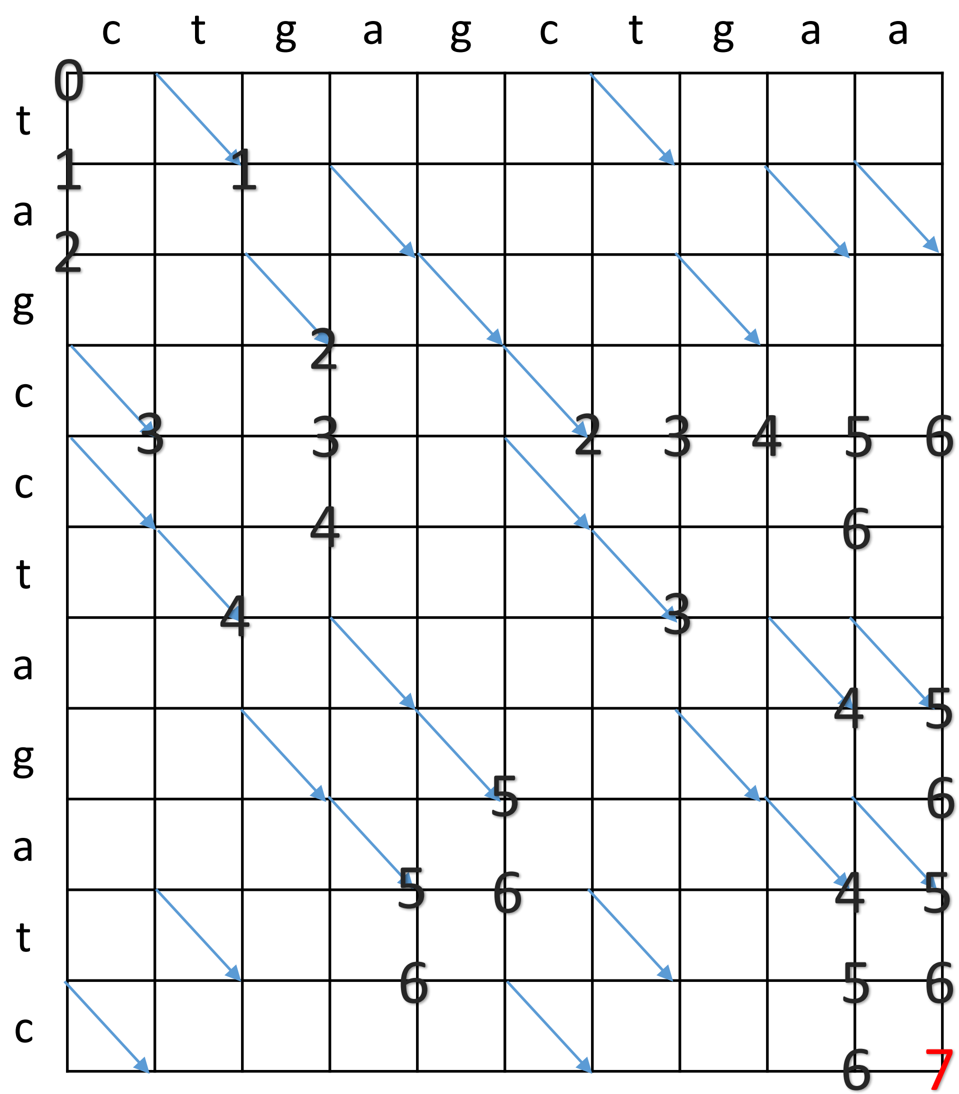

c t g a g c t g a a
t a g c c t a g a t c
0
1
1


c t g a g c t g a a
t a g c c t a g a t c
0
1
1
2
2
2

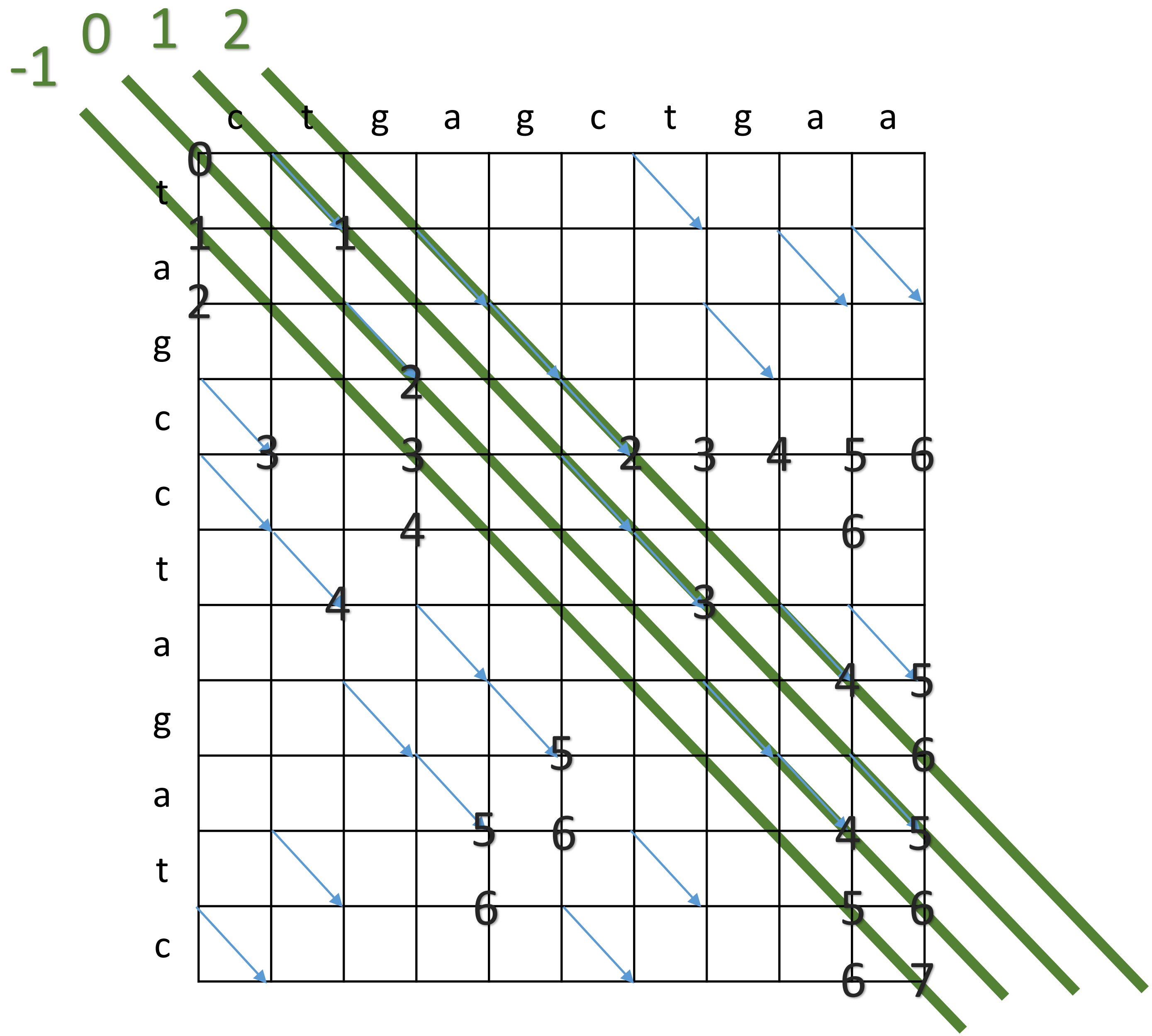


c t g a g c t g a a
t a g c c t a g a t c
0
1 1
2
2
3 3 2 3
3

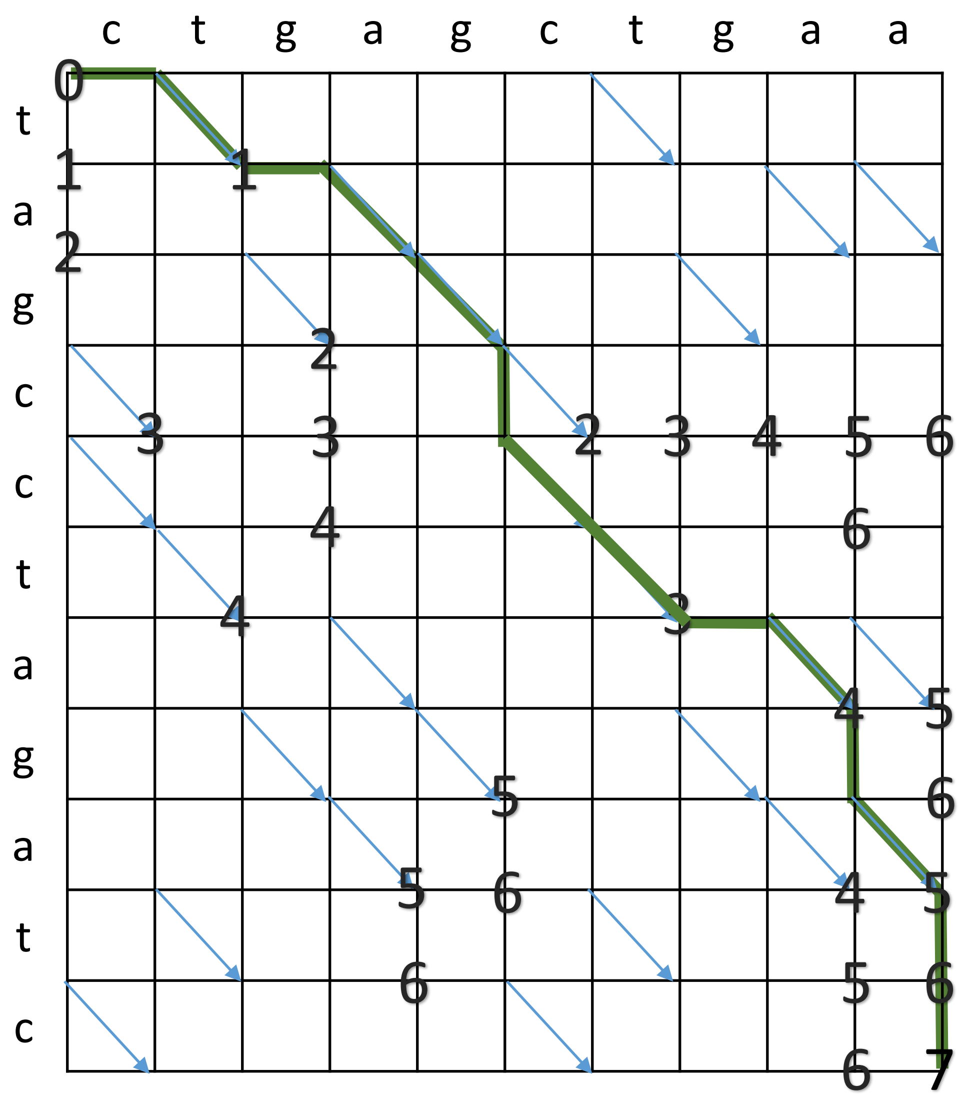

c t g a g c t g a a
t a g c c t a g a t c
0
1
1
2
2
2
3
3
3
4
4
4
4
3
4
4



c t g a g c t g a a
t a g c c t a g a t c
0
1 1
2
2
3 3 2 3 4 5 6
4 6
4 3
4 5
5 6
5 6 4 5
6 5 6
6 7

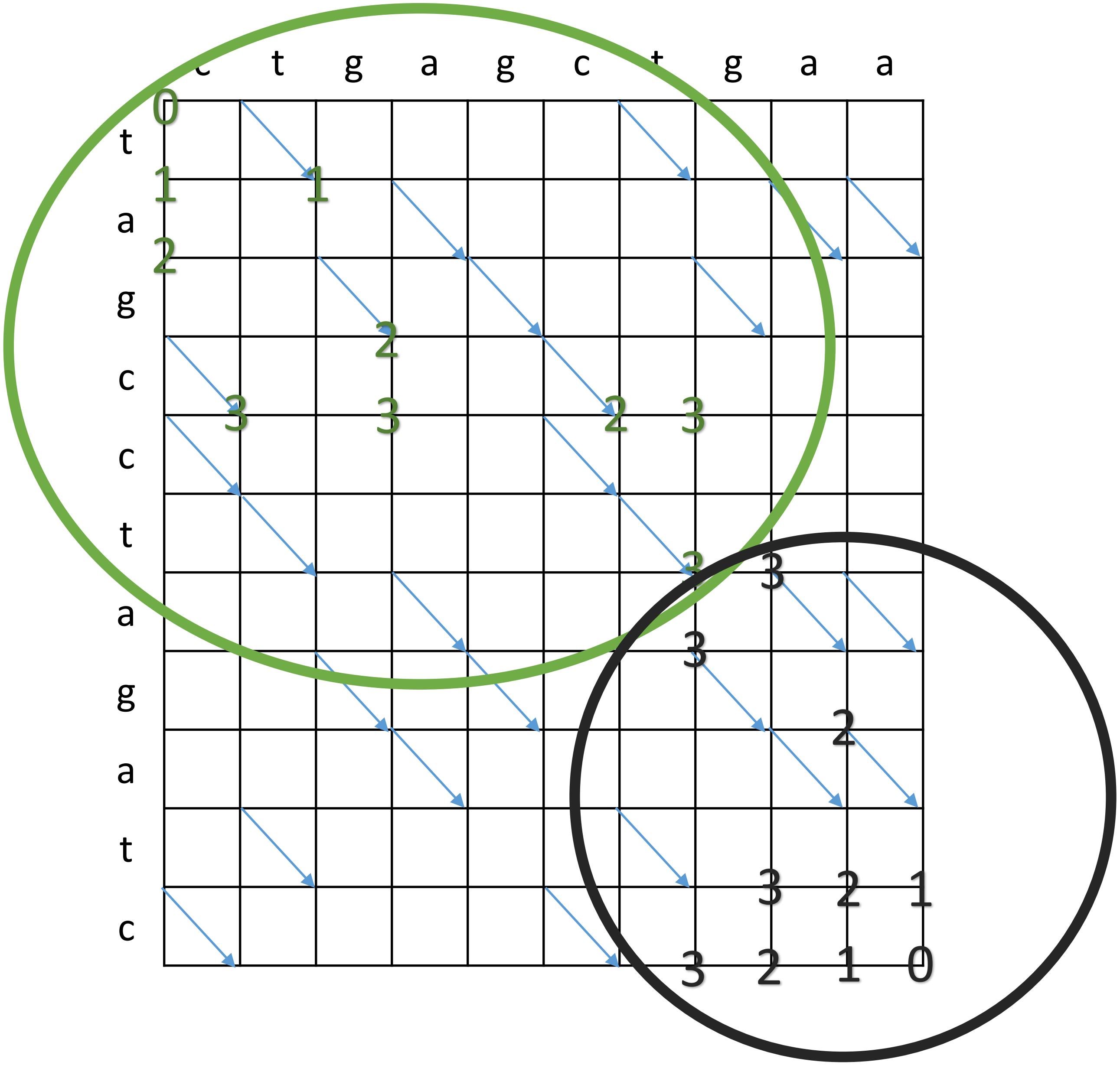


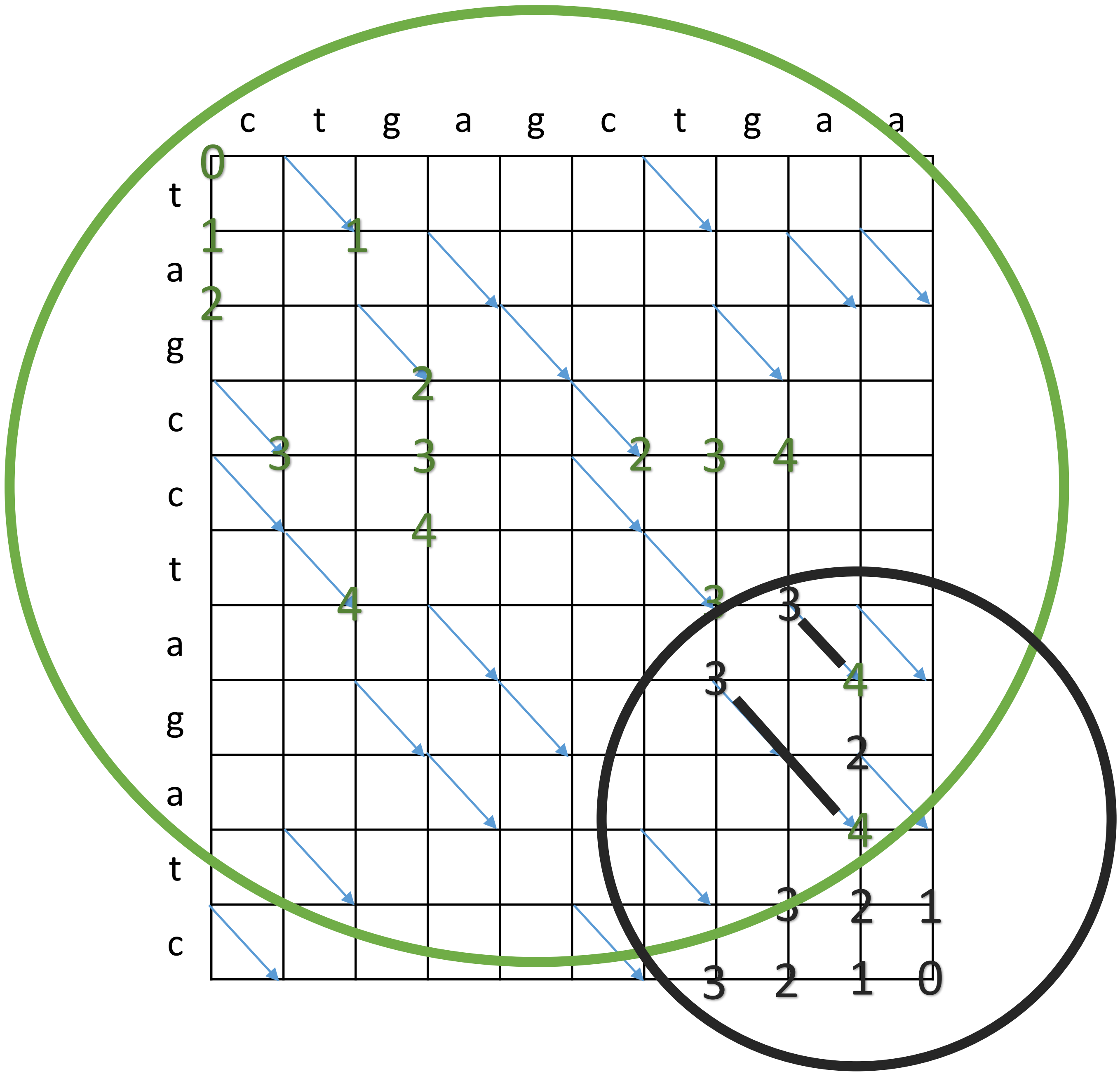
-1
0
1
2
c t g a g c t g a a
t a g c c t a g a t c

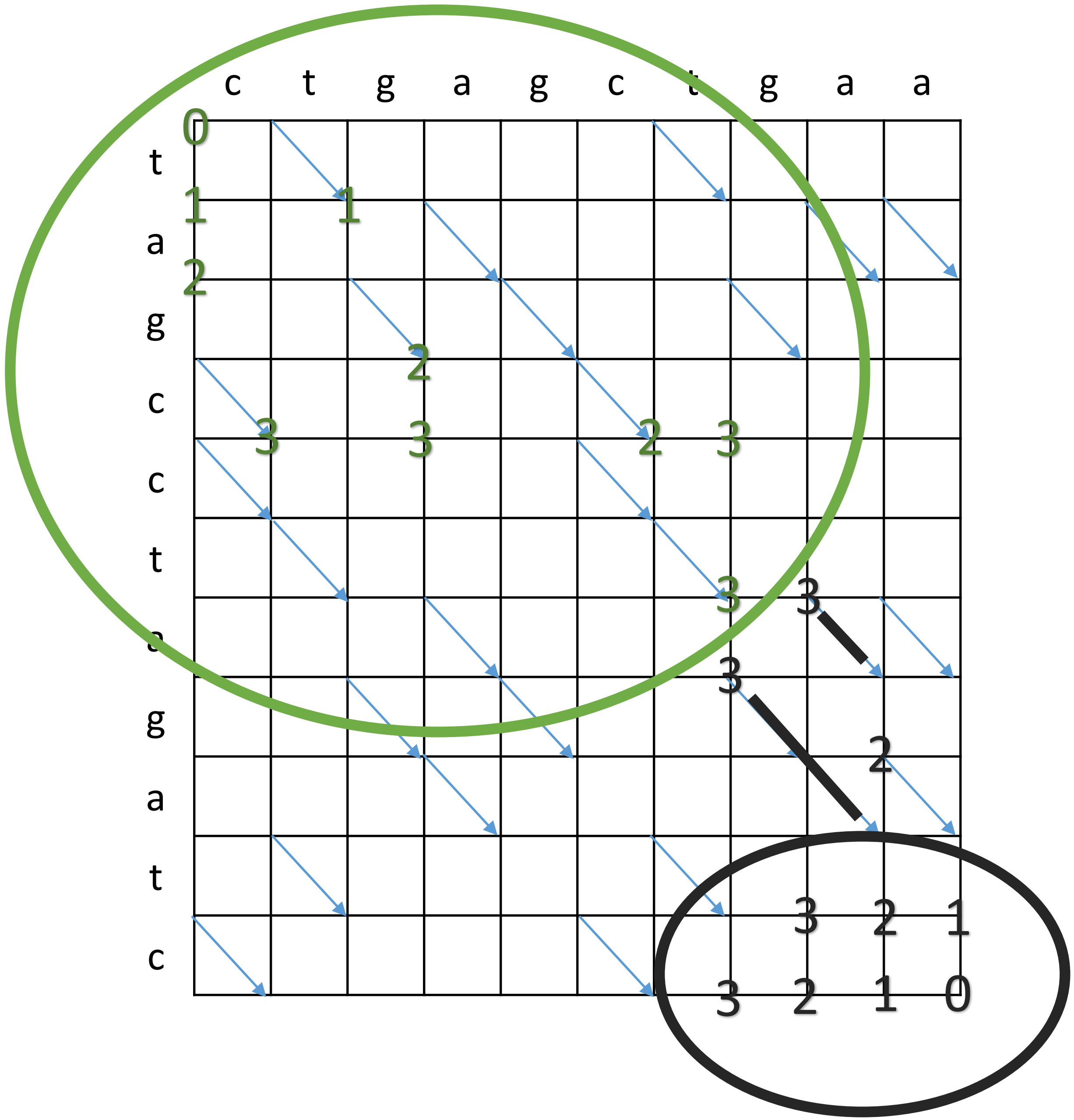


$O((M+N)D)$ time

$O(D^2)$ space


c t g a g c t g a a
t a g c c t a g a t c


c t g a g c t g a a
t a g c c t a g a t c
0 1 2 1 2 3 3 4 2 3 4 4
3 3 4 3 4 2 4
3 2 1
3 2 1 0



c t g a g c t g a a
t
a
g
c
c
t
a
g
a
t
c
0
1
1
2
2
3 3 2 3
3 3
3
2
3 2 1
3 2 1 0

```
LCS(A,N,B,M)
    If N>0 and M>0 Then
        Find the middle snake and length of an optimal path for A and B.
        Suppose it is from (x,y) to (u,v).
        If D > 1 Then
            LCS(A[1..x],x,B[1..y],y)
            Output A[x+1..u].
            LCS(A[u+1..N],N–u,B[v+1..M],M–v)
        Else If M > N Then
            Output A[1..N].
        Else
            Output B[1..M].
```

$$T(M+N,D) \le \begin{cases} \alpha(M+N)D + T\left(M1+N1,\frac{D}{2}\right) + T(M2+N2),\frac{D}{2}) \\ \beta(M+N) \end{cases}$$

$$\frac{D}{2} \le 2\frac{D}{3}, for\ D \ge 2$$

$$T(M+N) \le 3\alpha(M+N)D + \beta(M+N)$$

$O((M+N)D)$

time

$2O(D)+O(\lg D)+O(M+N)$

space